Career Opportunities for Microbiologists

Microbiologists are biological scientists who study about organisms that are generally so small and can only be seen with a microscope. These microorganisms include bacteria, algae, yeasts, fungi, protozoa, viruses, and other microscopic forms of life. Some microbiologists specialize in one type of microorganism. For example, bacteriologists concentrate on bacteria and virologists study viruses. Microbiologists isolate and make cultures of microorganisms, identify their characteristics, and observe their reactions to chemicals and other kinds of stimuli. They also study how microorganisms develop and reproduce as well as their distribution in nature.

The Scope of Microbiology (Course Benefit / Advantages)

The whole ecosystem depends on bacterial activities. The modern microbiology is a large discipline with different specialities. Microbiology has a great impact on fields such as medicine, agriculture, food sciences, ecology, genetics, biochemistry and molecular biology. There are many possible avenues of advancement for microbiologists.

Medical Microbiology – Medical microbiologists are involved in identifying the microorganisms causing the infectious diseases. They work on identifying the pathogens and assist the medical practitioners for prescribing the apt antibiotics in right dosages. They also study the ways in which the micro organisms cause the infection. Medical microbiologists study the relationship between microorganisms and disease establishment.

Immunology – Those who work on immune system related work are called immunologists. Immunologist study the body's defensive responses to microorganisms. They learn how our immune system protects our body during the infection. They suggest possible ways to increase our immunity. It is one of the fastest growing areas in science.

Microbial Ecology - The microbial interactions with living and non-living matters of the environmental habitats is referred to as microbial ecology. Microbial ecologists study the contributions of microorganisms to the cycling of various nutrients or elements. The ecologists are employed in reducing the pollution of the environment which is the burning issue in all metro cities. They work on employing microorganisms in bioremediation to reduce pollution effects.

Food and Dairy Microbiology – Some of our foods are actually the by-products of microbial growth. Example: Cheese is produced by the growth of microorganisms such as *Leuconostoc citrovorum* and *Streptococcus lactis*. Yoghurt results from the growth of bacteria such as *Lactobacillus bulgaricus* and *Streptococcus thermophilus* in milk. The leavening of bread is accomplished by *Saccharomyces cerevisiae* (Baker's yeast). Main work of the food and dairy microbiologists in the food industry is to prevent contamination during processing and the transmission

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